

VISHNU T PILLAI

Data Scientist | Machine Learning Engineer | AI Enthusiast

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with hands-on experience in Python, SQL, machine learning, deep learning, predictive modeling, and MLOps. Experienced in building end-to-end machine learning pipelines, including data preprocessing, feature engineering, model training, evaluation, deployment, and workflow automation. Skilled in Scikit-learn, XGBoost, TensorFlow, PyTorch, Pandas, NumPy, AWS, DVC, Feast, FastAPI, Flask, and Streamlit, with practical experience in computer vision, NLP, Generative AI, and Retrieval-Augmented Generation (RAG). Experienced in developing reproducible ML workflows using DVC, AWS S3, Git/GitHub, Docker, and CI/CD, and deploying machine learning and AI applications through REST APIs and interactive web applications.

EXPERIENCE

MLOps Engineer | Gofreelab Technologies

June 2025 – Present

- Developed an end-to-end MLOps pipeline for insurance cost prediction, covering data ingestion, data Preprocessing feature engineering, model training, evaluation, and workflow automation using Python.
- Implemented reproducible data versioning and feature management using DVC, AWS S3, and Feast, cutting manual data tracking effort by ~30% across 10+ training runs.
- Collaborated in an Agile team using Git/GitHub for version control, code reviews, and CI-driven testing/debugging, delivering well-documented, reproducible pipeline updates across sprints.

PROFESSIONAL CERTIFICATION

Data Science with Artificial Intelligence

ExpertzLab

- Completed professional training in Data Science, Machine Learning, Artificial Intelligence, Python, SQL, and Data Analytics, with hands-on experience in data preprocessing and exploratory data analysis.
- Applied Pandas, NumPy, Scikit-learn, Power BI, SQL, and Deep Learning to develop, evaluate, and deploy machine learning models through practical projects.

EDUCATION

Bachelor of Science in Computer Science (UNIVERSITY OF KERALA)

2025

PROJECTS

Insurance Cost Prediction | Machine Learning & MLOps Project

Present

- Developed an end-to-end regression pipeline for insurance cost prediction using Python, Scikit-learn, and XGBoost, achieving 94% R^2 — an 18% improvement over the baseline model.
- Built a production-oriented MLOps workflow covering data ingestion, preprocessing, feature engineering, model training, and evaluation, with reproducible pipeline versioning via DVC
- Deployed the trained model as a FastAPI REST API with an interactive Swagger/OpenAPI interface, enabling real-time insurance cost predictions via a live, publicly accessible endpoint
- LIVE DEMO API: [Insurance Cost Prediction API](#)

Hybrid RAG AI | Python, LangChain , ChromaDB , BM25

Present

- Built a RAG pipeline using LangChain, ChromaDB , BGE embeddings, and Groq LLM for document based Q&A.
- Implemented hybrid search with vector retrieval, BM25, and RRF for relevant context retrieval.
- Deployed a FastAPI + Streamlit application using Docker/Docker Compose with REST API and source citations.

SKILLS

• Technical Skills

- Programming & Databases: Python, SQL, MySQL
- Data Science: Pandas, NumPy, Exploratory Data Analysis (EDA), Predictive Modeling, Data Visualization
- Machine Learning: Supervised Learning, Unsupervised Learning, Scikit-learn Cross Validation, XGBoost
- Deep Learning: Neural Networks, TensorFlow, PyTorch, Computer Vision, Natural Language Processing
- Cloud & Deployment: AWS Cloud, Flask, FastAPI, Streamlit, Git, GitHub, Docker, DVC, CI/CD

• Soft Skills

- Critical Thinking, Problem Solving & Teamwork
- Communication & Data Storytelling
- Client Communication, Adaptability & Willingness to learn